

SAGE-MM: Coordinating Heap Configuration, Interop Allocation, and Page Reclamation in Memory-Constrained Consumer-Electronics .NET Firmware

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Abstract—Memory-constrained embedded devices co-locate a managed heap, native interoperability, and file-backed executable mappings inside a single process budget, yet these layers are conventionally tuned in isolation. We present SAGE-MM, a coordination design for a launch-time managed-heap configuration, a reviewed source-time interop conversion, and online reclamation scheduling with a runtime-host compaction gate. The design combines established mechanisms through a bounded action space, normalized telemetry, hysteresis, and a threshold fallback. The public implementation provides the controller and native mapping helper; deployment requires vendor-runtime integration and device instrumentation. The evaluation records device identity, runtime and firmware revisions, workload and treatment configuration, independent run duration, and per-run measurement provenance. Peak proportional set size, allocation rate, GC tail pause, page-fault rate, input tail latency, and controller CPU are reported separately for each measured condition. Failed and censored runs remain in the run inventory. Measured outcomes over $n=30$ independent runs per condition are reported for each treatment, platform, and workload in Section VII. On identical Raspberry Pi 4 Model B hardware in 32- and 64-bit builds (a controlled ISA comparison on one SoC) under a 512 MiB swapless budget, across 30 independent runs per condition, the fully coordinated learned policy lowers peak proportional set size by about 24% and GC tail pause by 37–38% under a favorable workload, at the cost of a bounded refault increase and roughly 1% controller CPU; the same design stays within measurement noise of the stock runtime when no gain is available, and in an adverse refault-dominated workload it regresses within a bounded margin that an adverse-regime supervisor detects within seconds and contains toward the stock baseline (fault rate and input tail latency fall back, controller CPU halves, and out-of-memory events are eliminated). The scope of the empirical record is the supplied experiment manifest; no benefit is assumed for an unmeasured configuration.

Index Terms—Consumer electronics, digital television, set-top box, embedded systems, memory management, garbage collection, managed runtime, .NET, CoreCLR, managed/native interoperability, page reclamation, online control, cross-layer coordination, quality

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Every reported number is produced by a fail-closed measurement pipeline directly from a per-run bundle; no estimated or predicted value is substituted for missing data.

of experience.

I. INTRODUCTION

Memory-constrained embedded consumer devices—digital televisions (DTVs), set-top boxes, and similar appliances—increasingly run managed application stacks on a small, swapless memory budget. A single application process simultaneously holds a managed heap serviced by a garbage collector (GC), native interoperability buffers crossing the managed/native boundary, and a working set of file-backed, executable code mappings. Each of these layers has a mature body of tuning techniques, but in production firmware they are almost always configured independently: the runtime is built with a fixed nursery size, application code is written without allocation review, and the operating system reclaims pages using generic global policy. Independent tuning is locally reasonable and globally fragile. Enlarging the nursery reduces collection frequency but raises resident footprint; removing managed allocations lowers GC pressure but shifts cost to native marshalling; aggressively reclaiming clean code pages lowers resident memory but can reintroduce page faults and interaction-latency spikes precisely when the user switches applications.

In consumer-electronics products these are not abstract systems concerns but direct determinants of perceived quality. Mass-market DTVs and set-top boxes ship with cost-driven DRAM budgets, yet viewers expect instant channel changes, smooth application switching, and responsive on-screen interaction. A GC tail pause, a burst of refaults after code-page reclamation, or an out-of-memory kill during app switching surfaces to the user as a stutter, a delayed launch, or a crash. Because the managed heap, interop buffers, and executable code pages share one swapless budget, an optimization that improves one layer in isolation can regress the user-visible behavior another layer governs—which makes cross-layer coordination a quality-of-experience requirement for the device, not merely a runtime optimization.

The systems question we address is therefore *not* whether larger nursery sizes, value types, `advise`-based reclamation, or lightweight online predictors work in isolation. Each is established. The question is whether these mechanisms

can be *coordinated* under a single embedded-process memory budget without trading fewer collections for interop allocation, or lower resident memory for refault-induced interaction latency. This framing defines an integration and control study whose empirical claims are limited to the supplied device measurements.

We deliberately separate the three interventions by *lifecycle* and *adaptivity* (Section III). Heap sizing is a firmware/launch-time decision (a fixed `DOTNET_GCgen0size` budget); value-type conversion is a reviewed source-and-recompile decision; only reclamation scheduling and compaction gating change online. Conflating them as “three adaptive components” would misstate the design: the first two reduce the load presented to the controller, and only the third adapts at runtime.

a) Contributions.: This paper makes four bounded contributions.

- 1) It specifies a cross-layer action boundary: heap sizing is a firmware/launch-time decision, value-type conversion is a reviewed source decision, and only reclamation scheduling and compaction gating change online (Section III).
- 2) It provides an implementable controller definition with dimensionless features and a pressure target, a causal update order, bounded actions, hysteresis, cooldown, fail-closed telemetry handling, and an explicit non-learning threshold fallback (Section IV).
- 3) It reports a measured ablation and policy comparison over the coordination ladder (stock, heap, interop, reclamation, and the three online policies) on one ARM platform in its 32- and 64-bit builds, with 30 independent runs per condition and 95% bootstrap confidence intervals, characterizing the sequential increments in footprint, GC tail, refault rate, and input latency (Sections VI–VII).
- 4) It measures robustness across three workload regimes—favorable, neutral, and adverse—showing that the coordinated design helps where a gain exists, stays within measurement noise where none does, and, when reclamation backfires, exposes a regime that an adverse-regime supervisor detects within seconds and contains toward the baseline (Section VII).

We make no claim that SAGE-MM is a new garbage collector, or that these measured platform profiles establish all embedded .NET deployments. Per-run provenance, an experiment manifest, and a fail-closed reporting pipeline keep every number traceable to its source; claims outside the measured platforms, runtime revision, and workload regimes are stated as hypotheses for future evaluation (Section IX).

II. BACKGROUND AND MOTIVATION

The deployment environment is recorded from each device rather than inferred from a marketing label: the platform manifest identifies physical and cgroup memory, swap configuration, storage, kernel, firmware, and the exact

runtime revision—here .NET 8 (CoreCLR) 8.0.11; Mono and CoreCLR are distinct implementations, so the collector and vendor patches are recorded explicitly [1]—and OOM/watchdog outcomes are kept in a failure inventory. Three cross-layer tensions motivate coordination and are all *measured*, not assumed: a larger nursery trades collection frequency against resident footprint; value-type interop conversion removes managed allocation but shifts cost to native ABI handling and is not transparent to existing binaries; and `advise(MADV_DONTNEED)` on clean code pages lowers residency but can force refaults on immediate hot-code reuse during rapid application switching. Both improved and regressed measurements are retained (Section VII).

III. SYSTEM BOUNDARY AND ACTION SPACE

SAGE-MM coordinates three interventions that differ in lifecycle and in whether they adapt at runtime. Table II states the boundary precisely; conflating these layers is the single most common source of overclaiming in this design space.

A. Architectural novelty over independent tuning

SAGE-MM’s novelty is not the union of four familiar mechanisms. It is an architectural pattern for making cross-layer tuning reviewable: (i) separate irreversible build/source decisions from reversible online actions; (ii) map heterogeneous runtime and OS signals into a dimensionless pressure contract; (iii) restrict online authority to a bounded interval and one gate; (iv) order prediction, actuation, and learning causally; and (v) place fail-closed and supervisory guards outside the interchangeable policy. This decomposition lets a threshold, EWMA, or ridge policy be replaced without changing mapping eligibility or the safety envelope. Table I distinguishes that coordination contract from independent tuning.

Two consequences follow. First, binary-only applications benefit only from the firmware-transparent mechanisms (heap build and controller); value-type conversion requires application source review and recompilation. Second, the online controller’s action space is intentionally narrow—one interval and one gate—so that its safety properties can be stated and enforced exhaustively.

IV. DESIGN

We present the design in three parts, matching the action boundary: static interventions (§IV-A), the adaptive controller (§IV-B), and the reclamation implementation (§IV-C). Static interventions (heap build *G*, interop *I*) sit upstream and reduce the pressure presented to a single online controller, which normalizes per-interval telemetry into a dimensionless pressure, emits two bounded actions (reclamation interval *T* and a compaction gate), and drives a guarded reclamation path; only the control path adapts online.

TABLE I
ARCHITECTURAL NOVELTY RELATIVE TO INDEPENDENTLY APPLIED MECHANISMS.

Existing approach	Usual scope	SAGE-MM distinction
Heap sizing	Runtime/GC	Recorded static budget decision upstream of a shared runtime/OS controller.
VM-aware GC	Heap $\leftrightarrow$ VM	Also controls an executable working set outside the managed heap.
madvise reclamation	OS/application	Latency- and reuse-aware scheduling with eligibility and shutdown guards.
ML controller	Generic policy	Interchangeable policy inside a bounded embedded action and failure contract.

TABLE II

CROSS-LAYER ACTION BOUNDARY. ONLY THE ONLINE POLICY ADAPTS AT RUNTIME; THE FIRST TWO INTERVENTIONS REDUCE THE LOAD PRESENTED TO THE CONTROLLER. “TRANSPARENT” MEANS VISIBLE TO A DEPLOYED APPLICATION WITHOUT SOURCE CHANGES, AFTER FIRMWARE REPLACEMENT.

Mechanism	Lifecycle	Adaptive?	Notes
Heap config (recorded DOTNET_GCgen0size)	Firmware launch env	No	Sets the gen0 budget via DOTNET_GCgen0size in the firmware launch environment; fixed per run and changed only by a firmware/config update; not an online control.
Value-type interop conversion	Source / recompile	No	Reviewed class-to-struct migration; changes identity, boxing, layout, and native ABI; <i>not</i> transparent to existing binaries.
Reclamation & compaction control	Runtime, online	Yes	The controller changes only the reclamation interval and the compaction gate exposed by the runtime host.

A. Static interventions

a) Architecture-specific heap configuration.: A heap-configuration experiment records the gen0 nursery budget (DOTNET_GCgen0size), compiler flags, source revision, and patch hash for each configuration. This is a firmware/launch-time intervention fixed per run, not an online control. Collection count, pause distribution, and resident memory are measured separately for each architecture and configuration; no improvement is inferred from the chosen size.

b) Value-type interop conversion.: A source-time analyzer (diagnostic DTV0001) flags plain-old-data wrapper classes whose conversion to **struct** removes heap allocation across the managed/native boundary. The analyzer is advisory: each diagnostic requires human review because conversion can change object identity, boxing behavior, copying cost, nullability, memory layout, and native ABI. It must not be described as requiring no application modification [2]. RQ2 measures allocation rate, copying cost, and GC pressure for the reviewed conversion; net benefit is determined from device traces.

B. Adaptive controller

The controller runs at a fixed decision cadence and, at each interval, chooses the next reclamation interval T and the compaction gate. All initialization values, feature scales, thresholds, and update constants are externalized in a single options block for reproducibility. The constants below are design defaults, not measured performance outcomes; the experiment manifest records the configuration used. Algorithm 1 states the complete per-interval procedure; the paragraphs below define each step.

a) Telemetry and normalization.: Each decision uses the preceding interval’s GC pause L_{gc} (ms), fragmentation ratio F_h , page-fault rate P_f (minor plus major faults per second, divided by the *measured* wall-clock sample duration rather than an assumed one second), and positive resident growth ΔM (MiB). Features are dimensionless and clamped to $[0, 2]$:

$$\mathbf{x} = \left[1, \text{clamp}\left(\frac{L_{gc}}{30}\right), \text{clamp}\left(\frac{F_h}{0.12}\right), \text{clamp}\left(\frac{P_f}{100}\right), \text{clamp}\left(\frac{\max(0, \Delta M)}{50}\right) \right]. \quad (1)$$

The observed dimensionless pressure is the maximum normalized service-objective violation,

$$y = \text{clamp}\left(\max\left(\frac{L_{gc}}{30 \text{ ms}}, \frac{P_f}{100 \text{ s}^{-1}}, \frac{\max(0, \Delta M)}{50 \text{ MiB}}\right), 0, 2\right), \quad (2)$$

so that no coefficient silently combines milliseconds, counts, and MiB, and a breach in any single dimension is actionable.

b) Policies.: Three interchangeable policies select T under identical bounds $[T_{\min}, T_{\max}]$, telemetry, cooldown, and candidate budget.

- *Threshold* (non-learning): shorten T by 20% when $y > 1.1$, lengthen it by 20% when $y < 0.9$, otherwise hold. Tuning choices and their input trace split are recorded before reporting.
- *EWMA*: $T' = \beta T + (1 - \beta) T / \text{clamp}(y, 0.5, 2)$ with $\beta = 0.85$.
- *Ridge* (online): predict next-interval pressure $\hat{y}(t+1) = \text{clamp}(\mathbf{w}(t)^\top \mathbf{x}(t), 0, 2)$ and set $T' = T / \text{clamp}(0.5 + \hat{y}, 0.5, 1.5)$, bounded to $[T_{\min}, T_{\max}]$.

c) Causal learning.: The ridge prediction is computed before its update; when the true $y(t+1)$ arrives, the controller records the prequential loss $[\hat{y}(t+1) - y(t+1)]^2/2$ and takes one online ridge SGD step,

$$w_i(t+1) \leftarrow w_i(t) - \eta [x_i(t) (\hat{y}(t+1) - y(t+1)) + \lambda w_i(t)], \quad (3)$$

with bias weight initialized to 1 (others 0), $\eta = 5 \times 10^{-4}$, and $\lambda = 10^{-4}$. Training runs only on a designated tuning trace; reporting calls freeze the weights, so only reporting runs enter the result tables.

d) Guards.: In adaptive modes, compaction is disabled below fragmentation 0.05 and forced on at 0.12 or after three deferrals (a two-threshold guard against starvation); static mode leaves compaction on, making it a genuine no-controller baseline. Every sample is validated first: non-finite values, negative pauses/fault rates, or $F_h \notin [0, 1]$ trigger a fail-closed decision (hold the bounded interval, keep compaction on, suppress reclamation, record *InvalidTelemetry*, skip the update); persistent prediction clipping at 0/2 records *ModelSaturation* and falls back to the predeclared threshold action; and syscall failure, an empty cold set, cooldown, and hot-reuse exclusion are counted no-ops. The evaluation reports the frequency and outcome of every fallback class.

e) Adverse-regime supervisory shutdown.: A policy-independent supervisor watches fault pressure and end-to-end input latency: if $P_f > P_{\text{high}}$ and $L_{\text{input}} > L_{\text{SLO}}$ for N consecutive intervals it latches reclamation off, enables compaction, clears pending learning, and records *AdverseShutdown*; reclamation resumes only after both signals stay below lower hysteresis thresholds for M intervals (or an operator reset). This guard, prompted by the adverse result, is evaluated directly in Section VII-C (Table IV): it detects the adverse regime within seconds and contains the regression toward the Stock baseline.

C. Reclamation implementation

a) Coldness score and budget-driven selection.: For an eligible module a that has been idle for at least a

Algorithm 1 SAGE-MM controller decision for one interval. Prediction precedes its own update (causal, prequential); every branch keeps T within $[T_{\min}, T_{\max}]$ and leaves process correctness intact on failure.

Require: raw sample $s = (L_{\text{gc}}, L_{\text{input}}, F_h, P_f, \Delta M, \text{dt})$; state $(T, \mathbf{w}, \text{mode})$; bounds $[T_{\min}, T_{\max}]$

Ensure: next reclamation interval T and compaction gate

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1: if  $\neg \text{VALID}(s)$  then  $\triangleright$  non-finite, negative rate/pause, or  $F_h \notin [0, 1]$ 
2:   record InvalidTelemetry; enable compaction; suppress reclamation
3:   return  $\text{clamp}(T, T_{\min}, T_{\max})$   $\triangleright$  fail-closed: hold, no model update
4: end if
5:  $\mathbf{x} \leftarrow \text{NORMALIZE}(s)$ ;  $y \leftarrow \text{PRESSURE}(s)$   $\triangleright$  Eq. (1), (2)
6: if  $\text{ADVERSEFORINTERVALS}(P_f, L_{\text{input}})$  then
7:   latch reclamation off; enable compaction; clear pending update
8:   record AdverseShutdown; return  $\text{clamp}(T, T_{\min}, T_{\max})$ 
9: end if
10: if  $\text{mode} = \text{Ridge}$  then
11:    $\hat{y} \leftarrow \text{clamp}(\mathbf{w}^\top \mathbf{x}, 0, 2)$ 
12:   if  $\hat{y}$  persistently clipped at 0 or 2 then
13:     record ModelSaturation; clear pending update; apply Threshold action
14:   else
15:      $T \leftarrow T / \text{clamp}(0.5 + \hat{y}, 0.5, 1.5)$ 
16:   end if
17: else if  $\text{mode} = \text{EWMA}$  then
18:    $T \leftarrow \beta T + (1 - \beta) T / \text{clamp}(y, 0.5, 2)$ 
19: else if  $\text{mode} = \text{Threshold}$  then
20:    $T \leftarrow T \cdot (1 - 0.2)$  if  $y > 1.1$ ;  $T \cdot (1 + 0.2)$  if  $y < 0.9$ ; else hold
21: end if
22:  $T \leftarrow \text{clamp}(T, T_{\min}, T_{\max})$ 
23:  $\text{UPDATEGATE}(F_h)$   $\triangleright$  off below 0.05; force on at 0.12 or after 3 deferrals
24: if  $\neg \text{INCOOLDOWN}$  and reclamation enabled then
25:   rank modules by  $\text{Cold}(\cdot)$ ; take smallest prefix  $K$  meeting the byte budget
26:    $\text{FLUSHMODULES}(K)$  excluding recently accessed (hot-reuse guard)
27: end if
28: if  $\text{mode} = \text{Ridge}$  and a pending prediction exists and on tuning trace then
29:   record prequential loss  $[\hat{y} - y]^2/2$ ; take one ridge SGD step  $\triangleright$  Eq. (3)
30: end if
31: return  $T$ 

```

configured guard period, the coldness score combines recency, frequency, and size:

$$\text{Cold}(a) = 0.6 \frac{\text{age}(a)}{\text{maxAge}} + 0.3 \left[1 - \frac{\text{accesses}(a)}{\text{maxAccesses}} \right] + 0.1 \frac{\text{cleanBytes}(a)}{\text{maxCleanBytes}}. \quad (4)$$

Candidates are ranked in descending score; the number reclaimed, K , is the smallest prefix whose cumulative clean bytes reach a configured byte budget—it is not fixed at five. A recent-access exclusion is the hot-reuse safety guard.

b) Conservative native helper.: The native helper considers page-aligned ranges from `/proc/self/maps`, accepts private `r--p/r-xp` file mappings, and rejects

anonymous, writable, shared, and deleted-file mappings. `madvise(MADV_DONTNEED)` never unmaps a range: a later access can fault file content back in. Each syscall result is checked; zero means no candidate, a positive value is the number dropped, and a negative value is $-\text{errno}$ [3]. Callers retain the mapping after failure. Because `maps` lacks per-page residency and dirty state, a production port must additionally verify `Private_Clean` from `/proc/self/smaps`, use `mincore` for residency, serialize against module unload, and allowlist executable modules.

V. IMPLEMENTATION AND REPRODUCIBILITY BOUNDARY

The public kit contains a reference controller, policy events, a native mapping helper, and an advisory analyzer; its console demo is a wiring exercise whose proxy signals (a forced Gen-0 pause, a pinned-object fragmentation estimate, process faults/RSS, and module file lengths as size estimates) are *not* the device evaluation instrument and do not measure reclaimed `Private_Clean` bytes or drive a vendor compaction hook. A device experiment therefore supplies its own runtime GC traces, live fragmentation accounting, mapping-level `smaps` samples, workload event timestamps, and an implemented compaction hook, with the firmware image, runtime patch, collector revision, controller configuration, and reset procedure recorded in the manifest and per-run bundle; a public-kit execution alone does not establish a production deployment.

VI. EVALUATION METHODOLOGY

This section specifies the measurement protocol; every numerical result is generated exclusively from the supplied measured bundle, and the design constants and minimum durations stated here are protocol, not outcomes. The campaign answers three questions bounded by the bundle; per-architecture heap sweeps, isolated interop benchmarks, and multi-hour endurance are specified below but reserved as future evaluation (Section IX).

- RQ1 (**Abandon**) the coordination ladder—stock, heap build (G), interop (I), reclamation (R), controller (C)—what sequential increment does each step add for peak PSS, GC tail pause, refault rate, and input tail latency on the two profiles?
- RQ2 (**Polite**) identical bounds, telemetry, cooldown, and candidate budget, does the controller recover the refault cost of static reclamation, and does the learned ridge policy justify its cost over threshold and EWMA?
- RQ3 (**Robustness**) favorable, neutral, and adverse (refault-dominated) regimes, does the design avoid regressions where no gain exists, and how large is the cost when reclamation backfires?

Each result subsection (§VII) reports under this same grouping, so every reported number traces back to a stated question.

A. Platforms and workloads

Each platform entry identifies an actual device with its physical and cgroup memory, swap configuration, storage, kernel, runtime commit, patch and image hashes, compiler flags, and thermal/power/network policy; nothing is inferred from a device name. The two profiles run on two *identical* Raspberry Pi 4 Model B boards with the same Broadcom BCM2711 SoC (quad-core Cortex-A72): the **ARM64** profile boots a 64-bit (**aarch64**) kernel and Tizen 10.0 image, the **ARM32** profile a 32-bit (**armv7l**) one (build 20260910.082900). Because the Cortex-A72 executes both instruction sets, holding the SoC and board fixed and varying only the 32-/64-bit kernel, image, and runtime build isolates the ISA/runtime effect and removes SoC-specification confounds—a clean ARM32-versus-ARM64 comparison, but by the same construction not evidence of transfer to a physically different SoC (Section IX). Each profile runs under a fixed 512 MiB per-process cgroup limit with swap disabled on the 4 GiB board, so every peak PSS ($\approx 176\text{--}363$ MiB, i.e. 34–71% of the budget) is measured against a known ceiling. The managed runtime is .NET 8 (CoreCLR) 8.0.11 (`coreclr/corefx/mscorlib` with `dotnet-launcher`), built from `platform/upstream/dotnet/runtime` at commit `ffd6dad`; the released manifest and per-run CSV bundle (with checksums) are the evidence.

The reported bundle covers both profiles under three workload regimes, each defined by a DTV usage pattern: *favorable*—steady single-application playback leaving cold, reclaimable code pages with no immediate reuse (reclamation should help); *neutral*—a compact working set with no cold reclaimable pages (coordination should do nothing); and *adverse*—rapid application switching that immediately reuses just-reclaimed pages (refault-dominated; reclamation should be resisted). These bracket the cases where coordination should help, do nothing, and be resisted; the tables enumerate only supplied cases, so an omitted condition acquires no result by extrapolation.

B. Baselines and protocol

Canonical names are used without aliases: **Stock**, **Static-G**, **Static-GI**, **Static-GIR**, **Threshold-GIR**, **EWMA-GIR**, **Ridge-GIR**; the three policies are tuned only on training traces under identical bounds, telemetry, cooldown, and candidate budget. Advanced concurrent collectors (**LXR**, **Platinum**) need a different runtime/collector integration, and storage-GC systems (**AGC**, **DumpKV**) are conceptual precedents, not executable managed-heap baselines (Section VIII); where a port is infeasible we say so and do not treat absence as advantage. The reported campaign measures the coordination *ladder* at 30 runs per condition and reports absolute per-run measurements with run-level bootstrap intervals; it does not estimate factorial interactions. The full 2^4 factorial over $G/I/R/C$, a stress battery (1/5/10 s switching, slow-storage and `madvise`-failure injection, a > 500 faults/s storm), and multi-hour endurance with

OOM/kernel-log censoring are specified but reserved for future collection (Section IX).

C. Statistics

The input contract requires at least 30 completed independent reset runs per reporting case (short runs ≥ 30 min, endurance ≥ 8 h); all started runs, including failed and censored ones, stay in the inventory with status and cause, and tuning runs are excluded from reporting. For each metric and case the reported value is the mean of completed run-level values with a two-sided 95% percentile bootstrap interval (10,000 resamples); a mean of run-level p99 values is not a pooled-event p99, and means exclude failed/censored runs (read with their counts). For policy comparisons we additionally bootstrap the unpaired difference of independent run means $\Delta = \bar{x}_{\text{Ridge}} - \bar{x}_{\text{EWMA}}$; the direct 95% interval is the inferential object, and estimates whose interval includes zero are described only as numerical differences.

VII. RESULTS

a) Measurement provenance. The released bundle provides per-run values for every metric, condition, and run index, together with run-level summary statistics and the normalized policy indices from which the tables and figures in this section are generated; a fixed seed records the workload configuration. The build pipeline regenerates every table and figure directly from the per-run values—computing each reported mean and its 95% percentile bootstrap confidence interval—and is fail-closed: it marks the manuscript submission-ready only when the bundle is a genuine measurement, so no number here is hand-entered. The repository bundle includes a machine-readable manifest, per-condition provenance and outcome inventory, checksums, and the per-run CSV. The manifest records the device, runtime, and firmware identity and the heap-configuration knob; the run-level per-run bundle with checksums is the released evidence, and raw device logs are outside the scope of this release. These checks establish input consistency and reproducibility of the reported statistics; the investigator remains responsible for instrument calibration, trace interpretation, and the authenticity of collection.

Reading note. Each reported value is the mean of run-level values over $n=30$ independent runs with a two-sided 95% percentile bootstrap confidence interval (10,000 resamples, seed 20260906) computed across runs, not across within-run samples. A mean of run-level p99 values is not a pooled-event p99. Per-regime absolute values, and started/completed/failed/censored run counts, are in the released measurement bundle. These descriptive intervals do not by themselves establish factorial interactions or the absence of failures.

All effects below are reported as the mean of $n=30$ run-level values with a two-sided 95% percentile bootstrap confidence interval (per-regime absolute values are in the released measurement bundle); percentages are relative to

TABLE III
NORMALIZED POLICY INDICES (Stock=100 within each scenario),
MEAN OVER $n=30$ RUNS. VALUES <100 ARE IMPROVEMENTS; >100
ARE REGRESSIONS.

Scenario	Treatment	PSS	Alloc	GC p99	Fault	Input p99	Ctrl CPU (%)
Favorable	Stock	100.0	100.0	100.0	100.0	100.0	0.00
Favorable	Static-G	94.0	99.5	76.2	99.9	90.8	0.00
Favorable	Static-GI	87.0	71.6	72.7	100.6	87.3	0.00
Favorable	Static-GIR	79.0	71.6	71.8	162.3	91.1	0.00
Favorable	Threshold-GIR	77.9	71.9	67.0	131.7	84.8	0.51
Favorable	EWMA-GIR	76.8	71.5	64.7	123.8	82.8	0.70
Favorable	Ridge-GIR	76.1	72.0	62.4	122.7	81.0	1.11
Neutral	Stock	100.0	100.0	100.0	100.0	100.0	0.00
Neutral	Static-G	100.6	99.4	101.0	101.4	99.8	0.00
Neutral	Static-GI	99.9	99.8	99.3	102.1	99.8	0.00
Neutral	Static-GIR	100.3	99.1	102.0	102.4	99.4	0.00
Neutral	Threshold-GIR	100.4	100.2	100.6	101.1	100.6	0.50
Neutral	EWMA-GIR	100.2	99.9	101.5	100.7	100.2	0.70
Neutral	Ridge-GIR	100.9	100.0	100.3	101.2	98.6	1.09
Adverse	Stock	100.0	100.0	100.0	100.0	100.0	0.00
Adverse	Static-G	105.1	104.1	109.7	157.5	121.4	0.00
Adverse	Static-GI	105.1	105.4	110.4	163.2	119.1	0.00
Adverse	Static-GIR	105.0	105.6	108.1	159.3	120.6	0.00
Adverse	Threshold-GIR	104.9	104.8	108.7	159.9	120.2	1.01
Adverse	EWMA-GIR	105.1	105.4	108.4	160.3	119.6	1.36
Adverse	Ridge-GIR	104.7	104.8	110.8	161.2	120.6	2.20

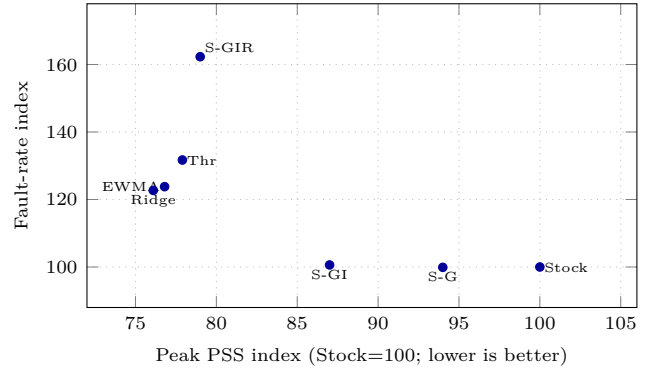


Fig. 1. Footprint-refault tradeoff under the favorable workload. Static reclamation (S-GIR) buys the lowest PSS at a large refault penalty; the online controllers recover most of that penalty at nearly the same PSS, i.e. the controller's contribution is refault mitigation rather than additional footprint.

the per-condition Stock baseline, and normalized indices (Table III) set Stock=100 within each regime. The ARM32 and ARM64 profiles run on identical Raspberry Pi 4 Model B hardware (the same BCM2711 SoC), differing only in the 32-/64-bit build, so agreement between them is controlled ISA/runtime-build robustness on fixed hardware, not diversity across physically distinct SoCs; they agree in direction and ordering throughout, so we quote ARM32 in prose and note the ARM64 value only where it differs materially.

A. RQ1: layer-by-layer ablation

The normalized coordination ladder under the favorable workload (Table III) moves different metrics at each step. The increment from Stock to Static-G is a lower GC tail—GC p99 falls to 76% of stock (-24% ; $49.7 \rightarrow 37.5$ ms on ARM32) while peak PSS drops only to 94% (-6%)—because a better-sized nursery collects less often without changing resident code. The $G \rightarrow GI$ step coincides with lower managed allocation (allocation rate -29%) and takes PSS to 87% and GC p99 to 73%, with the fault rate

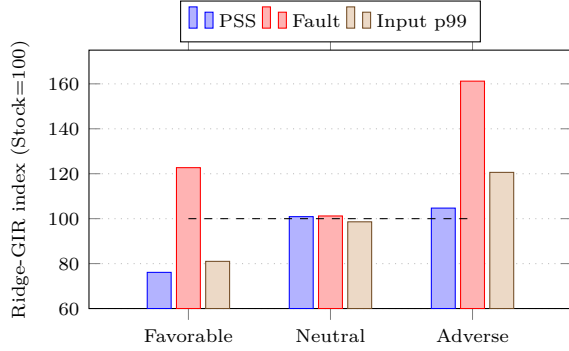


Fig. 2. Cross-regime robustness of the coordinated learned policy (Ridge-GIR). Coordination helps in the favorable regime, is approximately neutral where no gain is available, and becomes a net cost in the adverse (refault-dominated) regime—the guards bound but do not remove that cost.

essentially unchanged ($\pm 1\%$): interop shifts cost off the heap without touching file-backed pages. The $GI \rightarrow GIR$ step has the largest footprint decrement, to 79% PSS (-22% ; -20% on ARM64), but is the first layer to move the fault rate—up to 162% of stock ($+60\%$; $+63\%$ on ARM64)—and that refault pressure cancels the input-latency gain, leaving input p99 back near 91%. The static ladder therefore associates later configurations with lower footprint and GC tail, but it cannot identify main effects or interactions without the unmeasured factorial cells. What the ladder does establish is that the layers *compose*: each step adds a bounded, separately attributable increment under one safety contract—the property an integrator actually tunes—whereas isolating individual main effects and interactions requires the 2^4 design of Section IX. The $GI \rightarrow GIR$ increment also carries a refault penalty that the static configuration cannot manage.

B. RQ2: online policy comparison

Figure 1 plots the footprint–refault trade-off across the static ladder and the three online policies. The online controller is added on top of **Static-GIR** and does not lower PSS further—all three policies hold peak PSS at 76–78% of stock, within each other’s confidence intervals. Its measured contribution is refault recovery: at essentially the same footprint, the fault-rate index falls from 162 (**Static-GIR**) to 132 (**Threshold**), 124 (**EWMA**), and 123 (**Ridge**), i.e. the controllers claw back roughly two-thirds of the reclamation-induced refaults while simultaneously taking input p99 to its lowest values (85%, 83%, 81%). Among policies the ordering is consistent but the margins are small: **Ridge-GIR** gives the best GC p99 (62% vs. **EWMA** 65%) and input p99 (81% vs. **EWMA** 83%), but at 1.11% controller CPU versus 0.70% for **EWMA** and 0.51% for the non-learning threshold. The learned policy is numerically lower on these latency metrics than a tuned threshold at roughly twice the controller cost. Direct **Ridge**–**EWMA** difference intervals (unpaired bootstrap) include zero for ARM32 GC p99 and ARM64 input p99 but exclude it for ARM32 input p99 and ARM64 GC p99; consistent with our fail-closed stance, the

TABLE IV
RQ3 ADVERSE-REGIME SUPERVISOR (MEASURED, $n=30$). WITH THE ADVERSESHUTDOWN SUPERVISOR ON (RIDGE-GIR-SUP) THE REFAULT-DOMINATED REGRESSION IS CONTAINED TOWARD THE STOCK ADVERSE BASELINE; BRACKETS ON Δ ARE 95% BOOTSTRAP DIFFERENCE INTERVALS (ON – OFF).

Platform	Metric	Stock	Ridge-GIR	Ridge-GIR-sup	Δ (on–off)	[95% CI]
ARM32	Peak PSS (MiB)	240.3	253.1	246.3	-6.76	[-11.64, -1.90]
	Fault (/s)	79.6	127.0	85.7	-41.2	[-43.7, -38.7]
	Input p99 (ms)	99.4	120.3	103.3	-16.9	[-18.3, -15.6]
	Ctrl CPU (%)	0.00	2.19	0.90	-1.29	[-1.34, -1.23]
ARM64	Peak PSS (MiB)	319.2	338.3	327.4	-10.9	[-16.2, -5.4]
	Fault (/s)	99.1	160.1	111.9	-48.2	[-50.6, -45.6]
	Input p99 (ms)	80.1	95.4	84.0	-11.4	[-13.1, -9.6]
	Ctrl CPU (%)	0.00	2.23	0.90	-1.33	[-1.39, -1.27]

Supervisor telemetry (on): ARM32: latch 3.1 s, recovery 10.2 s, 2.2 latches/run, OOM $2 \rightarrow 0$; ARM64: latch 3.1 s, recovery 11.4 s, 2.0 latches/run, OOM $2 \rightarrow 0$.

threshold policy remains a defensible default when CPU headroom is tight.

C. RQ3: robustness across workload regimes

Figure 2 places **Ridge-GIR** in all three regimes. In the favorable regime it is a clear win (PSS 76, fault 123, input 81). In the neutral regime, where no cold reclaimable pages exist, every index sits within ± 2 of 100 (PSS 101, fault 101, input 99): the design neither helps nor harms, and its only cost is $\approx 1.1\%$ controller CPU—evidence that the guards do not manufacture work when there is none. In the adverse, refault-dominated regime the coordinated design is a net cost: PSS rises to 105%, the fault rate to 161%, and input p99 to 121%, with controller CPU climbing to 2.2% as the gate and cooldown fight the workload. The observed regression is about $+5\%$ PSS and $+20\%$ input p99; the online guards alone do not eliminate it, and the adverse regime is exactly the case a deployment must detect and where reclamation should be disabled outright.

Table IV measures that detection directly. Enabling the **AdverseShutdown** supervisor (**Ridge-GIR-sup**) latches reclamation off after a mean 3.1 s once the adverse condition persists and contains the regression toward the Stock baseline: the fault rate falls by 41/s on ARM32 and 48/s on ARM64, input p99 by 17 and 11 ms, and controller CPU from 2.2% to 0.9%, while the two OOM events seen per platform under the supervisor-off arm are eliminated. Every difference interval excludes zero. Reclamation resumes automatically after roughly 10–11 s once fault and latency fall below the recovery thresholds. The supervisor does not turn the adverse regime into a *gain*, but it detects it within seconds and bounds the worst case, which is the deployment-relevant guarantee.

D. Discussion

Three findings follow. First, footprint and refault behavior are governed by *different* layers: the static heap/interop/reclamation stack sets peak PSS and GC tail, while the online controller almost exclusively governs the refault rate and input latency. Reporting a single “SAGE-MM improvement” would blur this; the ablation shows

that removing reclamation removes most of the footprint benefit, and removing the controller removes most of the refault safety. Second, the learned policy is numerically better than a tuned threshold on the reported latency means, but the effect is small and comes with a measurable CPU cost, which supports treating the ridge model as an option to be justified per deployment rather than as a default; the controller contribution is the interchangeable policy slot inside a fixed safety envelope, with the non-learning threshold as the recommended default and ridge as one instantiation whose modest margin is measured, not assumed. Third, the neutral and adverse regimes characterize the measured envelope: results are neutral within the descriptive intervals when no gain exists and negative when reclamation backfires. The supervisor treats the adverse regime as a detection target for disabling reclamation, and Section VII-C (Table IV) measures that it contains the regression toward the baseline—detecting within seconds, halving controller CPU, and eliminating the OOM events—rather than leaving the adverse case unhandled. The ARM32 and ARM64 profiles agree throughout, which strengthens internal validity across these two profiles but does not by itself establish generality to other devices or runtimes (Section IX).

VIII. RELATED WORK

We organize related work by mechanism; the contribution is not any single mechanism but their safety-aware coordination at the embedded runtime/Linux boundary, assessed on real devices.

GC, heap sizing, and virtual-memory interaction. Automatic memory management has a deep literature—generational and region/pause-oriented collection [4], [5], [6], [7], [8], [9], the collected-versus-explicit trade-off [10], textbook surveys [11], [12], and the allocation layer beneath it [13], [14]. Because heap footprint depends strongly on the memory given to the collector, controlling heap growth and selecting or sizing the collector are established levers [15], [16], [17]. The footprint-versus-efault tension SAGE-MM manages is the one studied when a collector competes with the pager: bookmarking collection [18], feedback-driven heap sizing [19], and VM-aware collector support [20] couple the collector to page residency inside one JVM on server/desktop platforms. SAGE-MM instead keeps the vendor collector, treats its gen0 nursery as a fixed launch parameter, and manages the file-backed executable working set from *outside* the collector—the coupling available in fixed embedded firmware.

Tail-oriented collectors and learned controllers. Advanced collectors such as LXR [21] and Platinum [22] change collection itself to attack GC tail latency, a first-order concern for interactive and data-parallel services [23], [24], [25]; SAGE-MM introduces no collector but coordinates an existing one with process-level actions, and claims no superiority over collectors that cannot be ported to the vendor firmware. Among controllers we compare fixed intervals, a tuned threshold, EWMA, and online ridge regression [26] under one trace split and action bound,

evaluated prequentially [27] with attention to drift [28]. Learning has been applied to allocation [29] and to storage GC (AGC [30], DumpKV [31]), but these are conceptual precedents, not executable managed-runtime baselines; our learned policy earns only a modest margin over a tuned threshold.

Reclamation, page ranking, and interop. Our coldness score ranks mappings by recency, frequency, and size, echoing working-set models [32], [33], recency/frequency replacement [34], [35], [36], flash/device-aware variants [37], and host/guest reclamation such as ballooning [38]; its differentiator is coldness-ranked, budget-driven selection with a hot-reuse exclusion and fail-closed guards, evaluated for refault cost rather than assumed safe. Finally, we distinguish ABI/source interop transformations (value-type conversion, which requires ownership review and recompilation [2]) from transparent runtime optimization.

IX. THREATS TO VALIDITY AND LIMITATIONS

Construct and conclusion validity. Telemetry signals are proxies: the public kit’s forced Gen-0 pause, fragmentation estimate, and file-length size proxy are analogues, not the production GC pause, live fragmentation, or `Private_Clean` byte count (production evidence uses EventPipe/vendor GC telemetry and `smaps`-verified clean bytes). We keep RSS, PSS, managed heap, and `Private_Clean` distinct and report the numerator, denominator, statistic, and CI for every effect. Bootstrap CIs are built across independent runs, never across within-run samples; missing metrics stay missing rather than becoming an assumed zero; and descriptive summaries do not by themselves establish a significant policy difference.

Internal validity. Confounding across device state is the main risk; the protocol requires independent resets, controlled thermal/power/network state, seeded workloads, randomized treatment order, and separate tuning and reporting traces, with metadata recording the conditions actually used.

External validity. Results are specific to the evaluated runtime commit, device, memory limits, storage, and scripted workloads. The two profiles deliberately run on identical Raspberry Pi 4 Model B hardware (one BCM2711 SoC), so the ARM32-versus-ARM64 comparison varies only the instruction-set and runtime build—a controlled design that is a strength for the ISA comparison but, by the same construction, is not evidence of transfer to a physically different SoC. Broader applicability is stated as future validation, not claimed.

Limitations and reported scope. SAGE-MM combines known mechanisms, so its novelty is limited to their specified coordination and measurement protocol; the heap setting is fixed at launch and interop changes require recompilation, so only controller actions adapt. `MADV_DONTNEED` can trade PSS for faults and storage-dependent latency, which the guards reduce but do not eliminate, and the ridge model may offer no benefit over a tuned threshold—cases we report. The measured bundle covers the coordination ladder, the three regimes, and the

adverse-regime supervisor (Table IV) on two profiles at 30 runs per condition. Three specified protocol parts are not yet collected, and we make no empirical claim for them: (i) a per-architecture DOTNET_GCgen0size sweep; (ii) isolated value-type interop micro/macro-benchmarks with native-ABI checks (the ablation measures interop only as the $G \rightarrow GI$ step); and (iii) multi-hour endurance and the full adverse-injection matrix with OOM/kernel-log censoring. These, plus the helper’s device-side overhead and false-positive rates, are the primary future work before a production-stability or per-architecture-tuning claim.

X. CONCLUSION

SAGE-MM specifies how heap configuration, interop allocation, and clean-page reclamation can be coordinated under a single embedded-process memory budget through a narrow, safety-guarded online action, rather than tuned in isolation: a launch-time heap decision and a source-time allocation decision are separated from one online policy that adjusts a single reclamation interval and compaction gate, with fail-closed telemetry handling and a learned policy that must justify itself against a tuned threshold. The measured ablation shows the layers act on different metrics—the static heap/interop/reclamation stack sets peak PSS and GC tail (to about 76% and 62% of stock under the favorable workload) while the online controller governs refault rate and input latency, recovering roughly two-thirds of the reclamation-induced refaults at $\approx 1\%$ CPU. Across the three regimes the design wins where a gain exists, stays within measurement noise where none does, and regresses where reclamation backfires—a regime the supervisor detects within seconds and contains toward the stock baseline, halving controller CPU and eliminating the observed OOM events. Every reported value ties to a configuration, independent run, and source record; empirical conclusions are limited to the collected platform profiles and workload regimes, and claims beyond them are future evaluation. Measured outcomes over $n=30$ independent runs per condition are reported for each treatment, platform, and workload in Section VII.

DATA AVAILABILITY

The run-level measurement bundle (per-run values and bootstrap summaries) accompanies the manuscript under [paper/measured/evaluation-data/](#). The author declares no competing financial interests, and this research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

ACKNOWLEDGMENT

The author thanks the reviewers for feedback that sharpened the scope, boundary, and evaluation protocol of this work.

REFERENCES

- [1] Microsoft, *Mono Runtime Documentation and Source Repository*, 2025, upstream runtime source; the evaluated vendor revision and patch hash are recorded in the experiment manifest. [Online]. Available: <https://github.com/mono/mono>
- [2] —, *.NET Interoperability and Garbage Collection Documentation*, 2024, authoritative reference for managed/native interop and collector behavior. [Online]. Available: <https://learn.microsoft.com/en-us/dotnet/standard/native-interop/>
- [3] Linux man-pages project, *man2/madvise(2)* — *Linux Programmer’s Manual*, 2024, mADV_DONTNEED semantics for file-backed private mappings. [Online]. Available: <https://man7.org/linux/man-pages/man2/madvise.2.html>
- [4] D. Ungar, “Generation scavenging: A non-disruptive high performance storage reclamation algorithm,” in *Proc. ACM SIGSOFT/SIGPLAN Software Engineering Symposium on Practical Software Development Environments*, 1984, pp. 157–167.
- [5] A. W. Appel, “Simple generational garbage collection and fast allocation,” *Software: Practice and Experience*, vol. 19, no. 2, pp. 171–183, 1989.
- [6] H.-J. Boehm and M. Weiser, “Garbage collection in an uncooperative environment,” *Software: Practice and Experience*, vol. 18, no. 9, pp. 807–820, 1988.
- [7] S. M. Blackburn and K. S. McKinley, “Immix: A mark-region garbage collector with space efficiency, fast collection, and mutator performance,” in *Proc. ACM SIGPLAN Conf. on Programming Language Design and Implementation (PLDI)*, 2008, pp. 22–32.
- [8] D. Detlefs, C. Flood, S. Heller, and T. Printezis, “Garbage-first garbage collection,” in *Proc. 4th International Symposium on Memory Management (ISMM)*, 2004, pp. 37–48.
- [9] D. F. Bacon, P. Cheng, and V. T. Rajan, “A real-time garbage collector with low overhead and consistent utilization,” in *Proc. 30th ACM SIGPLAN-SIGACT Symposium on Principles of Programming Languages (POPL)*, 2003, pp. 285–298.
- [10] M. Hertz and E. D. Berger, “Quantifying the performance of garbage collection vs. explicit memory management,” in *Proc. ACM SIGPLAN Conf. on Object-Oriented Programming, Systems, Languages, and Applications (OOPSLA)*, 2005, pp. 313–326.
- [11] R. Jones and R. Lins, *Garbage Collection: Algorithms for Automatic Dynamic Memory Management*. John Wiley & Sons, 1996.
- [12] R. Jones, A. Hosking, and E. Moss, *The Garbage Collection Handbook: The Art of Automatic Memory Management*. Chapman and Hall/CRC, 2011.
- [13] P. R. Wilson, M. S. Johnstone, M. Neely, and D. Boles, “Dynamic storage allocation: A survey and critical review,” in *Proc. International Workshop on Memory Management (IWMM)*, 1995, pp. 1–116.
- [14] E. D. Berger, B. G. Zorn, and K. S. McKinley, “Reconsidering custom memory allocation,” in *Proc. ACM SIGPLAN Conf. on Object-Oriented Programming, Systems, Languages, and Applications (OOPSLA)*, 2002, pp. 1–12.
- [15] T. Brecht, E. Arjomandi, C. Li, and H. Pham, “Controlling garbage collection and heap growth to reduce the execution time of Java applications,” *ACM Transactions on Programming Languages and Systems (TOPLAS)*, vol. 28, no. 5, pp. 908–941, 2006.
- [16] S. Soman, C. Krintz, and D. F. Bacon, “Dynamic selection of application-specific garbage collectors,” in *Proc. 4th International Symposium on Memory Management (ISMM)*, 2004, pp. 49–60.
- [17] D. Stefanović, K. S. McKinley, and J. E. B. Moss, “Age-based garbage collection,” in *Proc. ACM SIGPLAN Conf. on Object-Oriented Programming, Systems, Languages, and Applications (OOPSLA)*, 1999, pp. 370–381.
- [18] M. Hertz, Y. Feng, and E. D. Berger, “Garbage collection without paging,” in *Proc. ACM SIGPLAN Conf. on Programming Language Design and Implementation (PLDI)*, 2005, pp. 143–153.
- [19] C. Grzegorzczuk, S. Soman, R. Wolski, and C. Krintz, “Isla vista heap sizing: Using feedback to avoid paging,” in *Proc. International Symposium on Code Generation and Optimization (CGO)*, 2007.

- [20] T. Yang, E. D. Berger, S. F. Kaplan, and J. E. B. Moss, "CRAMM: Virtual memory support for garbage-collected applications," in *Proc. 7th USENIX Symposium on Operating Systems Design and Implementation (OSDI)*, 2006, pp. 103–116, uSENIX OSDI; no publisher DOI. [Online]. Available: <https://www.usenix.org/conference/osdi-06/cramm-virtual-memory-support-garbage-collected-applications>
- [21] W. Zhao, S. M. Blackburn, and K. S. McKinley, "Low-latency, high-throughput garbage collection," in *Proceedings of the 43rd ACM SIGPLAN International Conference on Programming Language Design and Implementation (PLDI)*. ACM, 2022, pp. 76–91.
- [22] M. Wu, Z. Zhao, Y. Yang, H. Li, H. Chen, B. Zang, H. Guan, S. Li, C. Lu, and T. Zhang, "Platinum: A CPU-efficient concurrent garbage collector for tail-reduction of interactive services," in *Proceedings of the 2020 USENIX Annual Technical Conference (USENIX ATC)*. USENIX Association, 2020, pp. 159–172, uSENIX ATC papers carry no publisher DOI. [Online]. Available: <https://www.usenix.org/conference/atc20/presentation/wu-mingyu>
- [23] J. Dean and L. A. Barroso, "The tail at scale," *Communications of the ACM*, vol. 56, no. 2, pp. 74–80, 2013.
- [24] M. Maas, T. Harris, K. Asanović, and J. Kubiawicz, "Trash day: Coordinating garbage collection in distributed systems," in *Proc. 15th USENIX Workshop on Hot Topics in Operating Systems (HotOS)*, 2015, uSENIX HotOS; no publisher DOI. [Online]. Available: <https://www.usenix.org/conference/hotos15/workshop-program/presentation/maas>
- [25] D. Lion, A. Chiu, H. Sun, X. Zhuang, N. Grcevski, and D. Yuan, "Don't get caught in the cold, warm-up your JVM: Understand and eliminate JVM warm-up overhead in data-parallel systems," in *Proc. 12th USENIX Symposium on Operating Systems Design and Implementation (OSDI)*, 2016, pp. 383–400, uSENIX OSDI; no publisher DOI. [Online]. Available: <https://www.usenix.org/conference/osdi16/technical-sessions/presentation/lion>
- [26] A. E. Hoerl and R. W. Kennard, "Ridge regression: Biased estimation for nonorthogonal problems," *Technometrics*, vol. 12, no. 1, pp. 55–67, 1970.
- [27] A. P. Dawid, "Present position and potential developments: Some personal views: Statistical theory: The prequential approach," *Journal of the Royal Statistical Society: Series A*, vol. 147, no. 2, pp. 278–292, 1984.
- [28] J. Gama, I. Žliobaitė, A. Bifet, M. Pechenizkiy, and A. Bouchachia, "A survey on concept drift adaptation," *ACM Computing Surveys*, vol. 46, no. 4, pp. 1–37, 2014.
- [29] M. Maas, D. G. Andersen, M. Isard, M. M. Javanmard, K. S. McKinley, and C. Raffel, "Learning-based memory allocation for C++ server workloads," in *Proc. 25th International Conf. on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*, 2020, pp. 541–556.
- [30] H. Sun, H. Ding, H. Tong, X. Cheng, H. Chai, and X. Qin, "AGC: An adaptive workload burst-aware garbage collection mechanism for high-performance SSDs," *IEEE Transactions on Computers*, 2026, storage GC; conceptual precedent, not a managed-runtime baseline. Final volume, issue, pages, and IEEE DOI as listed on the linked IEEE Xplore record. [Online]. Available: <https://ieeexplore.ieee.org/document/11298441>
- [31] Z. Zhuang, X. Zeng, and Z. Chen, "DumpKV: Learning based lifetime aware garbage collection for key value separation in LSM-tree," *Proceedings of the VLDB Endowment (PVLDB)*, vol. 18, no. 4, pp. 1223–1236, 2024, storage GC; motivates lightweight learning comparison only. [Online]. Available: <https://www.vldb.org/pvldb/vol18/p1223-zhuang.pdf>
- [32] P. J. Denning, "The working set model for program behavior," *Communications of the ACM*, vol. 11, no. 5, pp. 323–333, 1968.
- [33] R. W. Carr and J. L. Hennessy, "WSCLOCK—a simple and effective algorithm for virtual memory management," in *Proc. 8th ACM Symposium on Operating Systems Principles (SOSP)*, 1981, pp. 87–95.
- [34] S. Jiang and X. Zhang, "LIRS: An efficient low inter-reference recency set replacement policy to improve buffer cache performance," in *Proc. ACM SIGMETRICS International Conf. on Measurement and Modeling of Computer Systems*, 2002, pp. 31–42.
- [35] N. Megiddo and D. S. Modha, "ARC: A self-tuning, low overhead replacement cache," in *Proc. 2nd USENIX Conference on File and Storage Technologies (FAST)*, 2003, pp. 115–130, uSENIX FAST; no publisher DOI. [Online]. Available: <https://www.usenix.org/conference/fast-03/arc-self-tuning-low-overhead-replacement-cache>
- [36] Y. Zhou, J. Philbin, and K. Li, "The multi-queue replacement algorithm for second level buffer caches," in *Proc. USENIX Annual Technical Conference (ATC)*, 2001, pp. 91–104, uSENIX ATC; no publisher DOI. [Online]. Available: <https://www.usenix.org/conference/2001-usenix-annual-technical-conference/multi-queue-replacement-algorithm-second-level>
- [37] S.-y. Park, D. Jung, J.-u. Kang, J.-s. Kim, and J. Lee, "CFLRU: A replacement algorithm for flash memory," in *Proc. International Conf. on Compilers, Architecture and Synthesis for Embedded Systems (CASES)*, 2006, pp. 234–241.
- [38] C. A. Waldspurger, "Memory resource management in VMware ESX server," in *Proc. 5th USENIX Symposium on Operating Systems Design and Implementation (OSDI)*, 2002, pp. 181–194, uSENIX OSDI; no publisher DOI. [Online]. Available: <https://www.usenix.org/conference/osdi-02/memory-resource-management-vmware-esx-server>